Open problems:
Physical vector spaces for quantum mechanics

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

From the brief

1. **Completeness of the Lie algebra.** *Research.* Definition 1 admits pathological cases: a Hilbert space with the defining observables of a qubit ("too many vectors") and a Schwartz space with position as the only defining observable ("too few observables"). "Addressing this gap is an open problem"; the brief expects a "final definition" once it is solved. (Brief, Sec. 5, "Completeness of the Lie algebra"; Sec. 6.)

2. **Lie algebras determine physical systems.** *Research.* "Two systems are physically equivalent if and only if their Lie algebras of observables are equivalent" is asserted, and is what makes QSTO:V consistent with [B]STDIST. (Brief, Theorem 5, proof; Log, "Differences", last item.)

3. **Ensembles on a quantum vector space.** *Research.* QSTO:V uses density operators ρ and $S(\rho) = -\operatorname{tr}(\rho \log \rho)$, and Sec. 5 calls the set of ensembles "the most natural space to consider", but no ensemble space over V is defined. In particular, whether the ensembles induce on the pure states the quotient topology of V, under which the pure states are second countable exactly when V is, is open. (Brief, Sec. 3, QSTO:V; Sec. 5.)

4. **Field theories.** *Research.* "For a field theory, we would expect the defining observables to be at least countable"; no field-theory example of a quantum vector space is given. (Brief, Sec. 5, "Number of defining observables".)

5. **Which observables generate unitary groups.** *Research.* Since symmetric and self-adjoint coincide on V, "not all self-adjoint operators generate a unitary transformation", and "to understand which ones the integrability of the system must be checked". A characterization, analogous to complete vector fields in Hamiltonian mechanics, is not given. (Brief, Sec. 5, "Symmetric vs self-adjoint".)

6. **Which changes of frame.** *Research.* The proof of Theorem 5 gets [B]WDQ from the premise that "the defining observables of A must be observables in B and vice-versa", which holds exactly when the two frames give the same topology on V (`topology_eq_iff_continuous`). The brief does not say which changes of frame satisfy it. For Example 3 the change of frame $y = \tan(\frac{\pi}{2} \operatorname{erf} x)$, which the brief "Unphysicality of Hilbert spaces for quantum mechanics" uses

against [B]WDQ for QSTO:H, does not: it takes a Gaussian out of $\mathcal{S}(\mathbb{R})$ (`SchwartzFrame.not_forall_frameUnitary_mem_range`), and the position of the second frame, carried back to the first, is not defined at that Gaussian (`SchwartzFrame.toLp_gaussSchwartz_not_mem_domain`). So QSTO:V gives [B]WDQ only for a class of changes of frame that map V onto itself, as Carcassi, Calderón and Aidala already require for Schwartz spaces (arXiv:2308.06669v3, Appendix, Sec. VIII). The class is not specified. The restriction cannot simply be dropped: a wave function on $\mathbb{R}$ with finite position variance in every smooth increasing coordinate has compact support, and then, unless it is zero, its Fourier transform does not (for the second step, `quantum_no_cutoff` in the brief "Unphysicality of Hilbert spaces for quantum mechanics"; the first is not formalized). Open: which group of changes of frame should be part of the data of a quantum vector space, and how it relates to the Lie algebra of the defining observables. For plain changes of variable $f \mapsto f \circ g$ on $\mathcal{S}(\mathbb{R})$, those that map $\mathcal{S}(\mathbb{R})$ into itself are characterized by Galbis and Jordá (Rev. Mat. Iberoam. 34 (2018), Thm. 2.3); the unitary changes of frame carry the extra factor $\sqrt{g'}$. (Brief, Theorem 5, proof; Log, "Problems found in the brief", last item.)

From the formalization

None. Definition 1 and Examples 3 (in n dimensions) and 4 are checked in full.